Year 9 Algebra Pre-Test

Diagnostic Assessment

Name:

____

Date:

____

Time Allowed: 20 minutes

Instructions: Answer all questions in the spaces provided. Show your working where applicable.

Part A: Patterns and Number Sense

1. Write the next number in this pattern:

2, 4, 6, 8, ______

2. Write the next two numbers in this pattern:

10, 20, 30, ______, ______

3. Draw the next shape in this pattern:

$\triangle$, $\circ$, $\circ$, $\triangle$, $\circ$, $\circ$, ______

4. Missing numbers: find the hidden value

$5 + \_\_\_\_\_\_ = 9$

$\_\_\_\_\_\_ - 2 = 8$

$12 = 8 + \_\_\_\_\_\_$

$3 \times \_\_\_\_\_\_ = 15$

5. True or False?

$4 + 4 = 5 + 3$
Circle one: **True** / **False**

$10 - 2 = 6 + 1$
Circle one: **True** / **False**

6. Making equations

Write two different numbers that add up to 10:

$\_\_\_\_\_\_ + \_\_\_\_\_\_ = 10$

Part B: Substitution

Given $a = 3$, $b = -2$, and $c = 5$, find the value of:

7. $2a + c$

8. $a^2 - 3b$

Part C: Simplifying Expressions

Simplify the following expressions completely:

9. $4x + 3y - x + 5y$

10. $3m \times 4mn$

11. $\dfrac{15a^2b}{5ab}$

Part D: Expanding and Factorising

12. Expand the brackets:

$4(2x - 3)$

13. Factorise fully (put into brackets):

$12y + 18$

Part E: Solving Equations

Solve the following equations to find the value of x:

14. $x + 7 = 15$

15. $3x - 4 = 11$

16. $5x + 2 = 2x + 14$

Part F: Applied Algebra

17. Plumber's Cost

A plumber charges a flat call-out fee of \$50, plus \$40 for every hour he works. Write an algebraic rule for the total cost (C) if he works for h hours.

18. Farm Animals

A farmer has a collection of chickens and pigs on her farm. In total, there are 20 animals. If you count all the legs, there are 56 legs in total.

How many chickens and how many pigs does the farmer have?

Teacher Answer Key

Part A: Patterns and Number Sense

1. 10
2. 40, 50
3. $\triangle$
4. Missing numbers: 4, 10, 4, 5
5. True/False:
   - $4 + 4 = 5 + 3$: **True**
   - $10 - 2 = 6 + 1$: **False**
6. Any valid pair, for example: $4 + 6 = 10$

Part B: Substitution

7. 11
   Working: $2(3) + 5 = 6 + 5 = 11$
8. 15
   Working: $3^2 - 3(-2) = 9 - (-6) = 9 + 6 = 15$

Part C: Simplifying Expressions

9. $3x + 8y$
10. $12m^2n$
11. $3a$
    Working: $\frac{15a^2b}{5ab} = \frac{15}{5} \times \frac{a^2}{a} \times \frac{b}{b} = 3 \times a \times 1 = 3a$

Part D: Expanding and Factorising

12. $8x - 12$
13. $6(2y + 3)$
    Working: The highest common factor of 12 and 18 is 6.

Part E: Solving Equations

14. $x = 8$
15. $x = 5$
    Working: $3x - 4 = 11 \implies 3x = 15 \implies x = 5$
16. $x = 4$
    Working: $5x + 2 = 2x + 14 \implies 3x = 12 \implies x = 4$

Part F: Applied Algebra

17. $C = 40h + 50$ (or $C = 50 + 40h$)
18. **12 chickens and 8 pigs.**
    Working: Let c be chickens and p be pigs.
    $c + p = 20$
    $2c + 4p = 56$
    From the first equation, $c = 20 - p$. Substitute into the second:
    $2(20 - p) + 4p = 56 \implies 40 - 2p + 4p = 56 \implies 2p = 16 \implies p = 8$.
    Then $c = 20 - 8 = 12$.